Milad Bafarassat

milad.bafarassat@gmail.com
Website LinkedIn GitHub Google Scholar Istanbul

Education

Sabanci University
BSc in Computer Science and Engineering, Double Major EE, Minor Physics, Istanbul, Turkey — *Sep 2021 – Jan 2026*
CGPA: 3.72/4.0 – Ranked 63/729 (top 8%) in Faculty of Engineering
Awarded the *Dilek Sabanci Scholarship* for exceptional academic performance.

Sharif University of Technology
BSc in Chemical Engineering & MBA Master's, Tehran, Iran — *Sep 2009 – Feb 2019*
Ranked **5th** in the national MBA entrance exam out of **30,000** applicants (Top 0.02%).
Ranked **1374th** in the nationwide undergraduate entrance exam out of **308,875** students (Top 0.4%).

Publications and Patents

[C1] C. Li, R. Yang, T. Li, **M. Bafarassat**, K. Sharifi, D. Bergemann, Z. Yang. (2024) *STRIDE: A Tool-Assisted LLM Agent Framework for Strategic and Interactive Decision-Making* **AutoRL Workshop, ICML 2024**

[C2] M. Bafarassat, M. Yazici, K. K. Tokgoz. (2025) *FET Modeling with Deep Neural Networks and GAN-Augmented Small Measurement Dataset* **SMACD 2025 (Oral Presentation)**

[P1] M. Bafarassat, K. K. Tokgoz. (2025) *A Deep Neural Network-Based FET Simulation Method and System Implementing the Model* **Patent Application No.:** 2025/008782 **Filing Date:** June 30, 2025 **Ref No.:** P7069-TR

Research Experience

Undergraduate Researcher, Sabanci University
AI for Wireless Communications, Advisor: Prof. Özgür Gürbüz — *Sep 2025 – Present*
Developing CORTEX, a transformer-based beam selection system for terahertz (THz) UAV communications. The architecture combines convolutional front-ends with causal transformers using rotary position embeddings (RoPE) for drone state prediction. Key techniques include a two-stage approach for angle prediction and beam selection, circular-aware angle refinement, knowledge distillation, and focal loss for class imbalance. *Manuscript in preparation for submission to the **IEEE Open Journal of the Communications Society** .*

Undergraduate Researcher, Sabanci University
Backdoor Attacks on Time-Series Models, Advisor: Prof. Emre Özfatura — *Sep 2025 – Present*
Investigating security vulnerabilities in state-of-the-art time-series models. Implementing and extending two classes of attacks: (1) input-aware poisoning methods that generate sample-specific triggers, and (2) generative-model-based trigger synthesis using GANs/conditional generators to create stealthy backdoors against RNNs, LSTMs, and transformer-based classifiers. Conducting threat-model design, empirical evaluations, and initial defense experiments.*Manuscript in preparation for submission to the **USENIX Security Symposium**.*

Remote Researcher, Shanghai AI Lab
Mechanistic Analysis of Language Models, Advisors: Dr. Jie Fu, Wenyu Du (HKU PhD) — *Jan 2025 – Mar 2025*
Researched mechanistic interpretability of transformer-based language models to improve AI reliability and safety. Reviewed state-of-the-art literature, analyzed transformer components, and implemented a Meta-CoT-inspired approach to automate mechanistic interpretability workflows.

Graduation Project, Sabanci University
FET Modeling using Deep Learning, Advisor: Prof. Korkut Kaan Tokgoz — *Feb 2024 – Jun 2025*
Developed robust FET models using data-driven techniques. Explored data augmentation via interpolation, GANs, and autoencoders to expand a limited dataset of 80,000 samples across 27 transistors. Trained deep learning models for device prediction tasks and integrated modeling workflows into SPICE/Verilog-A pipelines. Project resulted in a paper accepted for Oral Presentation at **SMACD 2025**.

Remote Researcher, Yale University
LLM Agent Framework for Strategic Decision-Making, Advisors: Prof. Zhuoran Yang, Dr. Li — *Oct 2023 – May 2024*
Conducted research on enhancing LLM-based multi-agent decision-making through tool integration. Contributed to a paper accepted to the ***AutoRL Workshop at ICML 2024*** and the ***INFORMS Workshop on Market Design at EC 2024***.

Undergraduate Researcher, Sabanci University
Determining Twitter Users' Political Affiliations using Computer Vision, Advisor: Prof. Onur Varol — *Jul 2023 – Oct 2023*
Implemented self-supervised clustering methods, including SwAV and Barlow Twins, to analyze more than 7 million profile pictures. Used cluster representations to estimate political affiliation distributions and human/bot ratios.

Remote Researcher, University of Michigan-Ann Arbor
Graph Neural Networks for Urban Traffic Prediction, Advisor: Dr. Rafegh Aghamohammadi — *Jan 2022 – Jun 2022*
Developed graph-based deep learning models to predict traffic flow patterns using Macroscopic Fundamental Diagram (MFD) principles. Represented urban road networks as spatial graphs and designed graph convolutional architectures to learn spatial dependencies, with temporal modules for joint multi-region flow-density forecasting. Explored transfer learning strategies to generalize models from data-rich regions to areas with limited sensor coverage.

Teaching Experience

Programming Fundamentals (CS201) — **Sabanci University**
Learning Assistant — *Spring 2024*

Internships

Intraverse — **Lugano, Switzerland (Remote)**
AI Engineering Intern — *Jun 2025 – Jul 2025*
Designed and implemented a middleware pipeline connecting Unity game engine events to LLM APIs, enabling dynamic NPC dialogue, decision-making, and skill systems. Developed a plugin-based architecture for extensible character and quest design, with blockchain-backed validation for NPC responses. Produced documentation, a functional prototype, and performance benchmarks for real-time multiplayer scenarios.

Work Experience

HezarDastan Holding — **Istanbul, Turkey (Remote)**
AI Product Manager — *Feb 2020 – Mar 2023*
Led a cross-functional team of 7 software engineers and ML researchers, and 2 operations and marketing team members. Co-founded and introduced 3 innovative AI-driven products, expanding the company's product portfolio. Spearheaded the development and implementation of **Recrupen**, an in-house Applicant Tracking System (ATS), streamlining hiring across 6 subsidiaries and processing 5000+ resumes annually to facilitate over 200 hires. Launched **QShoot**, an AI-based math problem-solving app, achieving 10K users within 2 months of launch. Launched **Gorgeous**, an AI-powered makeup recommendation app using user photos and event context; reached 1K users in 2 days.

HezarDastan Holding — **Tehran, Iran**
Human Resources Manager — *Sep 2015 – Jan 2020*
Managed a team of 40 professionals across HR, Recruitment, Learning & Development, and Administration. Built and scaled a recruitment team from scratch to 7 full-time recruiters. Drove company growth by increasing headcount from 100 to 300 within two years, with a focus on technical roles. Promoted to HR Manager within 3 years due to consistent performance and leadership. Improved employee satisfaction significantly, raising the Net Promoter Score (NPS) from low 80%s to 92%.

Extracurricular Activities

Program for Undergraduate Research (PURE)
Mentor — *Jan 2025 – Jun 2025*
Mentored junior students on a project focused on leveraging reinforcement learning techniques to enhance transistor data augmentation and optimize learning outcomes.

Sabanci AI Club (kAi)
Student Advisor — *Apr 2023 – Oct 2023*
Helped in the identification and negotiation processes with potential sponsors for the club.

Select Coursework

Graduate-level courses: Internet of Things Sensing System, Scalable Learning Systems

Advanced courses: Probability and Statistics (A), Linear Algebra (A), Algorithms and Data Structures (A-), Quantum Mechanics I (A-), Quantum Mechanics II (B+), Intro to Signal Processing (A), Advanced Programming (A), Computer Networks (B+), Multimedia Communication (A), Information and Coding Theory (B+), Statistical Modeling (A), Data Science (A), Computer Vision (A-), Digital Image & Video Analysis (A-)

Skills

Software Skills: C++, Python, PyTorch, TensorFlow
Languages: English (C1), Turkish (C1), Azeri (Native), Persian (Native)

Tests and Scores

GRE (323): Quantitative: 170 (perfect score, 27/27 correct), Verbal: 153
TOEFL iBT (100): Reading: 24, Listening: 29, Speaking: 21, Writing: 26